- The r's complement of N is $r^n - N$ and the complement of $(r^n - N)$ is $r^n - (r^n - N) = N$. (The complement of the complement restores the number to its original value.

1.5.3 Subtraction with r's Complements

The subtraction of two positive numbers (M−N), both of base r, may be done as follows:

1. Add the minuend M to the r's complement of the subtrahend N.
2. Inspect the result obtained in step 1 for an end carry:
   (a) If an end carry occurs, discard it.
   (b) If an end carry does not occur, take the r's complement of the number obtained in step 1 and place a negative sign in front.

The following examples illustrate the procedure:

Example 1.5: Using 10's complement, subtract 72532−3250

M = 72532
N = 03250

```
10's complement of N =   100000          72532
                        - 03250        + 96750
                        -------        -------
                          96750         169282
                                        ↳ end carry.
```

answer: 69282 □

Example-1.6: Subtract: $(3250 - 72532)_{10}$

M = 03250
N = 72532

```
10's complement of N =   100000          03250
                        - 72532        + 27468
                        -------        -------
                          27468          30718
```

No carry   10's complement of 30718:

```
  100000
 - 30718
 -------
   69282
```

answer: −69282

Example-1.7: Use 2's complement to perform M−N with the given binary numbers.

(a) M = 1010100
N = 1000100

```
2's complement of N:   0111011          1010100
                             1        + 0111100
                       -------        ---------
                       0111100         10010000
```

end carry: 1
answer: 10000